

it can be changed.

She reasons that ‘we need to look at the factors we can control’—our thinking and behaviour. ‘When a behaviour, which is within the “normal” range, is needed to be changed, we can look at changing thinking and perception, which reinforce it’. (This also suggests that when behaviour is outside the normal range, such as addiction, it may not be as easily changed. We’ll look at this towards the end of Part Three.)

One of the behaviours that researchers believe we can easily change to affect our sensory system or inherent behaviour is how we pay for our goods or services—cash versus non-cash payments, such as card, smartphone or ‘buy now pay later’ platforms. In a 2019 study called ‘Cash, Card or Smartphone: The neural correlates of payment methods’, five Italian scientists examined whether our brain activity could be modulated by the method of payment used.

In their paper, the researchers acknowledged that for hundreds of years the use of coins and banknotes have engaged and stimulated our sensory system. What they wanted to understand was whether we received the same stimulation from digitised payments—in this case through payments with cards or smart phones. Researchers discovered a far greater brain activation in the insula or pain region of the brain when they were showing payments with cash and coins. In other words, the person experienced pain and had a negative reaction. Far less brain activation was detected when the same individual was watching a payment by card or by smartphone. In other words, it hurt less. The supposition is that electronic and digitised payments would cause even less pain.

Why does pain matter? This research suggests that paying by cards and other digitised payments, ‘likely mitigates the emotions linked to reward and regret around accumulating or paying money’, which adversely affects our spending behaviour. We experience less regret, so we spend more and in turn save less.

This means that, while we may be individually naturally inclined to behave a certain way, modern society is causing us to be desensitised to spending and saving patterns due to the reduction of pain associated with using digitised payments. Consumer research consistently supports these findings. In 2001, Drazen Prelec and Duncan Simester of MIT found